

Unit 1, Lesson 3: Applying the Laws of Exponents – Part 2

Benchmark

MA.7.NSO.1.1 Know and apply the Laws of Exponents to evaluate numerical expressions and generate equivalent numerical expressions, limited to whole-number exponents and rational number bases.

Learning Targets

- ☐ I can use the quotient of powers, power of a quotient, and zero exponent laws to simplify expressions.
- ☐ I can rewrite expressions using the laws of exponents to generate equivalent expressions.

1.3.1 Warm-Up: Notice and Wonder

In May 2021, the world's longest suspended pedestrian bridge in the world opened in Arouca, Portugal. The bridge runs over 1700 feet across the canyon, nearly 600 feet above the Paiva River.



1. What do you notice?
2. What do you wonder?

1.3.2 Exploration: Exponent Laws

1. Evaluate each of the following expressions.

$\frac{10}{10} =$

$\frac{-7}{-7} =$

$\frac{\frac{1}{2}}{\frac{1}{2}} =$

2. If the numerator and denominator have the same value, the quotient is _____.

3. The **Quotient of Powers Exponent Law** can be used to rewrite expressions when dividing two powers that have the same nonzero base.

a. Complete the table.

Quotient	Expanded form	Exponential Form a^b
$\frac{5^5}{5^2}$	$\frac{5 \cdot 5 \cdot 5 \cdot 5 \cdot 5}{5 \cdot 5}$	$5^{\square}$
$\frac{(-5)^7}{(-5)^4}$		
$\frac{(2)^6}{(2)^3}$		

b. How do the exponents in the first column relate to the exponents in the third column?

c. Work with your partner to complete the rule for finding the quotient of powers.

$\frac{a^m}{a^n} =$

☐

 $a^{mn}.$

☐

 $a^{m-n}.$

☐

 $a^{m+n}.$

4. The **Power of a Quotient Exponent Law** can be used to rewrite expressions when a fraction is raised to an exponent.

a. Complete the table.

Expression	Expanded Form	Exponential Form a^b
$\left(\frac{2}{5}\right)^4$	$\frac{2}{5} \cdot \frac{2}{5} \cdot \frac{2}{5} \cdot \frac{2}{5}$	$\frac{2^4}{5^4}$
$\left(-\frac{4}{7}\right)^3$		

b. How do the exponents in the first column relate to the exponents in the third column?

- c. Work with your partner to complete the rule for finding the power of quotients.

$\left(\frac{a}{b}\right)^m =$

☐ $a^{b-m}.$

☐ $\frac{a^m}{b^m}.$

☐ $a^m - b^m.$

5. The **Zero Exponent Law** is used to rewrite expressions when a nonzero rational number is raised to the zero power.

- a. Complete the table.

Quotient	Quotient of Powers Law	Expanded Form	Simplified Form
$\frac{4^5}{4^5}$	$4^{5-5} = 4^{\boxed{}}$	4 4 4 4 4 4 4 4 4 4	
$\frac{(-5.7)^2}{(-5.7)^2}$			
$\frac{\left(\frac{1}{2}\right)^6}{\left(\frac{1}{2}\right)^6}$			

- b. Complete the statement.

Any nonzero rational number raised to the 0 power is equal to _____.

1.3.3 Bring It Together: Exponent Laws



Quotient of Powers Exponent Law

$$\frac{a^m}{a^n} = a^{m-n} \text{ and } a^{m-n} = \frac{a^m}{a^n}$$



Power of a Quotient Exponent Law

$$\left(\frac{a}{b}\right)^m = \frac{a^m}{b^m} \text{ and } \frac{a^m}{b^m} = \left(\frac{a}{b}\right)^m$$



Zero Exponent Law

$$a^0 = 1$$

1.3.4 Your Turn!

1. Complete the table by rewriting each expression as an equivalent expression using the laws of exponents.

Exponential Expression	Equivalent Expression	Exponent Law(s) Applied
$\frac{(13)^6}{(13)^3}$		☐ product of powers ☐ quotient of powers ☐ power of a product ☐ power of a quotient ☐ power of a power ☐ zero exponent
$\left(\frac{3}{-5}\right)^5$		☐ product of powers ☐ quotient of powers ☐ power of a product ☐ power of a quotient ☐ power of a power ☐ zero exponent
$(-3)^2 \cdot (-3)$		☐ product of powers ☐ quotient of powers ☐ power of a product ☐ power of a quotient ☐ power of a power ☐ zero exponent
$(-4)^0$		☐ product of powers ☐ quotient of powers ☐ power of a product ☐ power of a quotient ☐ power of a power ☐ zero exponent
$\left(-\frac{3}{15}\right)^{15}$		☐ product of powers ☐ quotient of powers ☐ power of a product ☐ power of a quotient ☐ power of a power ☐ zero exponent

$\frac{(-7.45)^{11}}{(-7.45)^{11}}$		☐ product of powers ☐ quotient of powers ☐ power of a product ☐ power of a quotient ☐ power of a power ☐ zero exponent
$\frac{9^9}{9^4}$		☐ product of powers ☐ quotient of powers ☐ power of a product ☐ power of a quotient ☐ power of a power ☐ zero exponent

1.3.5 Collaboration: Analyzing Equivalent Expressions

Noah and Elena each generated a list of equivalent expressions by applying exponents laws.

Work with your partner to analyze each list for errors.

1. Noah generated the following expressions equivalent to 5^9 .

$\frac{5^{18}}{5^2}$	$(5 \cdot 5 \cdot 5 \cdot 5 \cdot 5 \cdot 5 \cdot 5 \cdot 5)$	$\left(\frac{-10}{-2}\right)^9$
$\frac{5^{12}}{5^3}$	$5^7 \cdot 5^2$	$\left(\frac{1}{5}\right)^9$

Circle the equivalent expressions and draw an X through the expressions that are not equivalent to the original expression.

2. Elena generated the following expressions as equivalent to $(2.95)^0$.

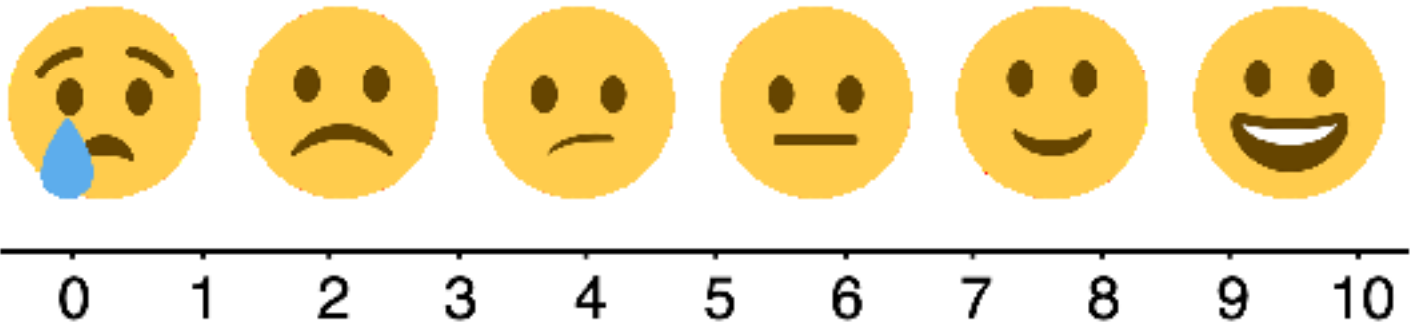
1	0	10^0
$\frac{2.95^9}{2.95^9}$	2.95	$\frac{\left(\frac{2}{3}\right)^{49}}{\left(\frac{2}{3}\right)^{49}}$

Circle the equivalent expressions and draw an X through the expressions that are not equivalent to the original expression.

1.3.6 Wrap-Up: Reflection

Circle the value on the scale that matches your understanding of each learning target.

1. I can use the quotient of powers, power of a quotient, and zero exponent laws to simplify expressions.



2. I can rewrite expressions using the laws of exponents to generate equivalent expressions.

